

EXERCISE - 3

Google App Engine SDK and Web App Development

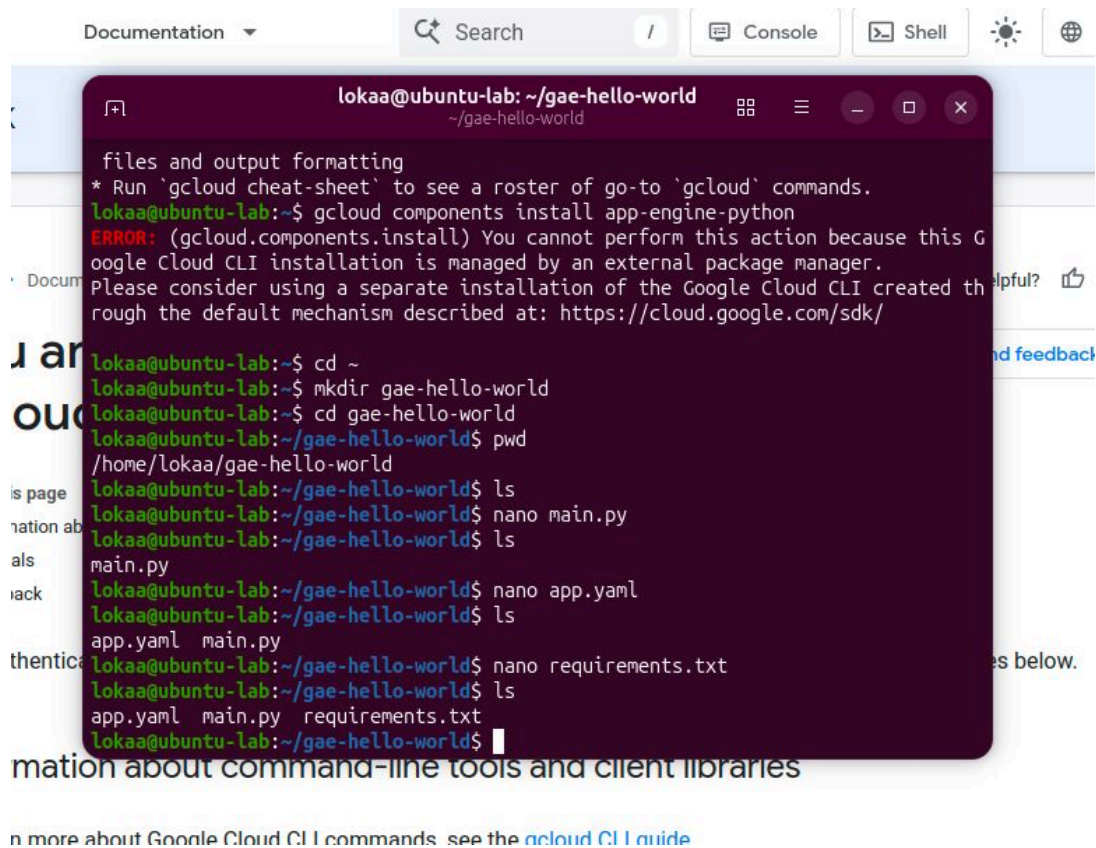
Aim:

To install and configure the Google Cloud CLI and develop a simple Hello World web application using Python Flask for Google App Engine.

Algorithm:

1. Start the Ubuntu virtual machine and open the Terminal.
2. Install and configure the Google Cloud CLI.
3. Initialize Google Cloud and authenticate using a Google account.
4. Select or create a Google Cloud project.
5. Create a directory for the Google App Engine application.
6. Create the Python application file using the Flask framework.
7. Define a route to display the Hello World message.
8. Create the App Engine configuration file and specify the Python runtime.
9. Create the requirements file containing the required Flask dependency.
10. Verify that the required application files have been created successfully.

Output:



The screenshot shows a terminal window titled 'lokaa@ubuntu-lab: ~/gae-hello-world'. The user is following a tutorial to set up a Python application for Google App Engine. The terminal output shows the user running 'gcloud components install app-engine-python', which results in an error: 'ERROR: (gcloud.components.install) You cannot perform this action because this Google Cloud CLI installation is managed by an external package manager. Please consider using a separate installation of the Google Cloud CLI created through the default mechanism described at: https://cloud.google.com/sdk/'. The user then navigates to the project directory, creates a 'gae-hello-world' directory, and enters it. They list the files, showing 'main.py'. They then create 'app.yaml' and 'requirements.txt' files using the 'nano' editor. The terminal output shows the user running 'nano app.yaml', 'nano requirements.txt', and finally listing the files, showing 'app.yaml main.py requirements.txt'.

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files and output formatting
* Run 'gcloud cheat-sheet' to see a roster of go-to 'gcloud' commands.
lokaa@ubuntu-lab:~$ gcloud components install app-engine-python
ERROR: (gcloud.components.install) You cannot perform this action because this Google Cloud CLI installation is managed by an external package manager. Please consider using a separate installation of the Google Cloud CLI created through the default mechanism described at: https://cloud.google.com/sdk/

lokaa@ubuntu-lab:~$ cd ~
lokaa@ubuntu-lab:~$ mkdir gae-hello-world
lokaa@ubuntu-lab:~$ cd gae-hello-world
lokaa@ubuntu-lab:~/gae-hello-world$ pwd
/home/lokaa/gae-hello-world
lokaa@ubuntu-lab:~/gae-hello-world$ ls
main.py
lokaa@ubuntu-lab:~/gae-hello-world$ nano main.py
lokaa@ubuntu-lab:~/gae-hello-world$ ls
app.yaml main.py
lokaa@ubuntu-lab:~/gae-hello-world$ nano requirements.txt
lokaa@ubuntu-lab:~/gae-hello-world$ ls
app.yaml main.py requirements.txt
lokaa@ubuntu-lab:~/gae-hello-world$
```

Result:

Thus, the **Google Cloud CLI** was successfully configured and a **Python Flask Hello World** application for Google App Engine was created successfully.